

From Maxwell's Rod to Unruh's Thermometer

Rethinking the Basis of Dimensional Analysis after Relativity

Lee McCulloch-James

Department of Natural Sciences, Dwight School London

Abstract

Dimensional analysis rests on three base dimensions, mass, length and time, that James Clerk Maxwell fixed in 1873. The method that uses them was formalised by Buckingham and Bridgman decades later, after special relativity had shown that length and time are not quantities all observers agree on. The base was never revisited. This essay asks what a dimensional basis chosen after Einstein would look like. It keeps only what relativity leaves invariant: the spacetime interval, which fixes the speed of light and merges length with time, and proper acceleration, which survives as a covariant scalar where velocity does not. Adding temperature, measured as energy, closes a two-quantity basis in which mass is energy and length-time is inverse acceleration. The Unruh effect supplies the constant relating acceleration to thermal energy, and a short dimensional argument shows that constant must be Planck's; the same effect explains why the two quantities are canonically paired without being made linearly dependent. The exercise reframes Boltzmann's constant as a unit convention and locates dimensional analysis, usually taken for bookkeeping, as a question about which quantities relativity permits us to call fundamental.

1 A basis older than relativity

WHEN a student first balances the dimensions of an equation, checking that force really does come out as mass times length over time squared, they are using a scheme that James Clerk Maxwell set down in the opening chapter of his *Treatise on Electricity and Magnetism* in 1873 [1]. Maxwell argued that all physical quantities can be expressed in terms of three fundamentals, which he took to be mass, length and time. A rod fixes the length, a clock the time, a prototype the mass, and every other quantity follows as a product of powers of these three. The scheme has served for a century and a half, and it works.

There is a curiosity in its history that is rarely remarked. Maxwell wrote in 1873, before relativity, so his treatment of length and time as absolute standards, the same for every observer, was the only treatment available to him. But the mathematical machinery that turned his scheme into the modern tool, Buckingham's π theorem of 1914 [2] and Bridgman's lectures of 1922 [3], was written after Einstein. By the time dimensional analysis was formalised, special relativity had been in the literature for nine and then seventeen years, and it had shown that a length is contracted and a time dilated by the Lorentz factor $\gamma = (1 - v^2/c^2)^{-1/2}$, so that neither the metre nor the second is a quantity on which observers in relative motion agree.

Historical observation

Maxwell chose M, L, T in 1873. Buckingham (1914) and Bridgman (1922) formalised the method after Einstein, and kept the same base. The choice is pre-relativistic; the formalism that fixed it is not.

Bridgman, who thought harder than most about what it means to define a physical quantity by the operations used to measure it, might have been expected to notice that two of his three base quantities had just lost their observer-independence. He did not revisit them, and

neither has anyone since. The γ that governs the disagreement between observers never enters a dimensional analysis; we write $[L] = L$ and $[T] = T$ as though 1905 carried no lesson for the choice of base. The results stay correct, because dimensional analysis tracks the structure of a unit and not its behaviour under a change of frame. Yet the base has been inherited from quantities that relativity marks as belonging to the observer rather than to the world.

Why are we still using Maxwell's 1873 basis after Einstein?

The question this essay pursues is a simple one. If we were to choose a dimensional basis today, keeping only what relativity certifies as invariant, what would it contain? The answer turns out to be shorter than Maxwell's list, and to place at its centre a quantity Maxwell never considered fundamental.

2 What relativity leaves standing

Special relativity removes the absoluteness of length and time, but in the same stroke it supplies quantities that all inertial observers agree on, and these are the natural candidates for a modern base.

The first is the spacetime interval. Where two observers disagree about a spatial separation dx and a temporal one dt , they agree on the combination

$$ds^2 = c^2 dt^2 - dx^2,$$

which is the same in every inertial frame. Taking the interval as the fundamental unit block, in place of length and time as separate standards, is the dimensional expression of what relativity actually asserts. In practice this is the familiar step of setting $c \equiv 1$: because the speed of light is the one velocity every observer shares, fixing it declares that space and time are measured in a single unit, related along the light-cone. Length and time cease to be two dimensions and become one, which we may write X , with $[L] = [T] = X$.

Key consequence

Fixing $c \equiv 1$ is the statement that the interval, not the rod or the clock, is the invariant, and it reduces the mechanical base from three dimensions to two.

The second surviving invariant is more surprising, and it is the pivot of the whole construction. Velocity is exactly the quantity the Lorentz transformation alters; there is no absolute velocity, which is the content of the principle of relativity itself. Position and duration are frame-dependent for the same reason. But *proper acceleration*, the acceleration an observer reads on an accelerometer carried in their own frame, is a covariant scalar. It is the magnitude of the four-acceleration, and every observer computes the same value for it. In a theory that has stripped away absolute velocity, absolute length and absolute time, absolute acceleration remains.

That acceleration should outrank velocity as a fundamental quantity is counter-intuitive, because our pre-relativistic instincts treat velocity as the simpler notion. Relativity inverts the ranking. Uniform velocity is undetectable, indeed meaningless without a reference frame, whereas acceleration is felt, measured, and agreed upon. A basis chosen for invariance rather than familiarity puts acceleration in and leaves velocity out.

3 Closing the basis with temperature

Two quantities remain to be settled: the source of mass, and the treatment of temperature. They turn out to be the same question.

Merging length and time leaves the mechanical world two-dimensional, mass and the combined X . Acceleration supplies one of these, since with $c = 1$ its dimension is that of inverse length, $[A] = X^{-1}$. Mass has still to come from somewhere, and no combination of acceleration and the interval will produce it, because neither carries mass. The missing ingredient arrives with temperature, provided temperature is understood correctly.

Among invariant quantities, thermal energy is distinguished because quantum field theory already identifies it canonically with proper acceleration through the Unruh effect.

A temperature is, physically, an energy per degree of freedom; the kelvin is a historical accident of having measured heat before recognising it as molecular motion. Measured in its natural currency, temperature is an energy, with dimension ML^2T^{-2} . Written this way it carries a factor of mass, and that is the factor the basis was missing. Taking the thermal quantity Θ with the dimension of energy, and working with $c = 1$, the two base quantities (A, Θ) close the mechanical world. Acceleration has dimension X^{-1} and thermal energy has dimension M , so

$$\mathbf{D} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \det \mathbf{D} = 1, \quad \begin{aligned} M &= \Theta, \\ X &= A^{-1}. \end{aligned}$$

Mass is energy, and length-time is inverse acceleration. The entire mechanical world is generated by two invariant quantities once the separate standard of length, the last of Maxwell's three absolutes, is given up.

Key consequence

In the post-relativistic basis, mass is energy ($M = \Theta$) and length-time is inverse acceleration ($X = A^{-1}$). Two invariants replace Maxwell's three.

This treatment of temperature has an immediate corollary. If the thermal quantity is an energy from the start, then Boltzmann's constant, whose job is to convert kelvin into joules, has nothing to convert. It does not appear in the basis. This is the natural endpoint of a view now enshrined in the SI, which since 2019 has defined the kelvin by fixing the numerical value of k_B [5]: a defined conversion factor rather than a measured constant. Measuring temperature directly in energy completes the move, and k_B is absent rather than set to one.

4 The effect of the Unruh Effect

With the spacetime interval already adopted as the single spacetime dimension X , so that $[L] = [T] = X$ and $c \equiv 1$, proper acceleration has dimension $[A] = X^{-1}$. The second basis quantity is the thermal energy $\Theta \equiv k_B T$, which carries the remaining mass dimension. The basis is therefore (A, Θ) , while M and X are derived through $M = \Theta$, $X = A^{-1}$. An observer with proper acceleration a in the Minkowski vacuum registers the Unruh temperature [4],

$$T = \frac{\hbar a}{2\pi k_B c}. \quad (1)$$

Multiplying by k_B and using the thermal basis quantity $\Theta \equiv k_B T$, this becomes

$$\Theta = \frac{\hbar a}{2\pi c}. \quad (2)$$

Boltzmann's constant has therefore not been discarded after appearing in the law. It has already been incorporated into the definition of the thermal coordinate Θ . Equation (2) also makes the role of Planck's constant transparent. In the interval basis, acceleration has dimension X^{-1} , whereas thermal energy has dimension M . A relation between them therefore requires a conversion factor of dimension MX . Restoring c for dimensional clarity,

$$[\hbar/c] = \frac{ML^2T^{-1}}{LT^{-1}} = ML = MX.$$

Thus the factor $\hbar/c$ is not an additional assumption introduced by the Unruh effect being rather the dimensional bridge required to convert inverse spacetime interval into energy. Once $c \equiv 1$, and $\hbar \equiv 1$ is also adopted, the correspondence Eq. (2) reduces to takes its fully natural-unit form,

$$\Theta = \frac{a}{2\pi}.$$

The dimensions therefore determine the appearance and power of $\hbar$; the Unruh calculation supplies the dimensionless coefficient $1/2\pi$.

Key consequence

Writing the thermal basis quantity explicitly as $\Theta \equiv k_B T$ gives

$$\Theta = \frac{\hbar a}{2\pi c}.$$

In the interval basis, a has dimension X^{-1} and Θ has dimension M , so a factor of dimension MX , $\hbar/c$ is required. Dimensional analysis fixes the bridge; the Unruh effect fixes the coefficient $1/2\pi$.

5 A pairing, not a dependence

A question now presses. If a law of physics relates the two base quantities, are they really independent? In linear algebra, two vectors related by $\mathbf{v}_1 = \lambda \mathbf{v}_2$ are dependent, and cannot both belong to a basis. Does Unruh collapse (A, Θ) in the same way?

Dimensional analysis has always answered such questions differently from linear algebra, and the difference is instructive. The metre and the second are treated as independent base dimensions throughout physics, even though the speed of light provides a universal conversion between them. Kelvin and joule were held independent for a century despite the existence of k_B . A conversion constant preserves the independence of two units because it is an empirical constant and not a definition. The metre is related to the second by a measured c ; it is not defined as a second times a number.

The decisive question is therefore whether the Unruh relation is a universal conversion of that kind. It is not. The relation $L = cT$ holds for every length and every time, without exception, which is what allows c to serve as a conversion while metre and second remain independent. Equation (1) holds only for uniformly accelerated observers in the Minkowski vacuum. A hot gas at rest has temperature and no acceleration; an accelerating spacecraft has acceleration and a wholly negligible Unruh temperature. Across the range of physical situations the correspondence between acceleration and temperature is many-to-many, an identity confined to one special class of states.

Acceleration and thermal energy are independent invariant quantities that become canonically identified through Unruh effect.

The right description of the Unruh relation is that it identifies a distinguished correspondence between two independent quantities, in the way that $E = hf$ identifies a distinguished correspondence between energy and frequency with Planck's constant fixing a canonical pairing between them, holding wherever quantum mechanics applies. In just the same way, $k_B T = \hbar a / 2\pi c$ fixes a canonical pairing between thermal energy and proper acceleration, holding on the submanifold of accelerated vacuum states. The two quantities remain independent, and the dimensional closure of the previous section stands. The basis is not constrained by a law of nature, any more than the SI metre-second basis is constrained by the existence of light.

6 The cold vacuum basis

One feature of the construction deserves to be drawn out, because it is the clearest expression of what the whole scheme is built on. The temperature in the Unruh relation is observer-relative. For a uniformly accelerated observer it takes the value in Eq. (1); for an inertial observer in the same vacuum it is exactly zero. The vacuum is cold to anyone in free fall and warm only in proportion to proper acceleration.

The single thermodynamic quantity in the basis therefore vanishes for inertial observers and, for accelerated ones, measures precisely the single kinematic invariant the theory retains. A basis assembled from relativity's invariants ends by tying its one thermal quantity to its one kinematic quantity, through the vacuum itself. Where Maxwell's scheme rested on three absolutes, this one rests on two invariants, the interval and the acceleration, with energy standing in for mass and the Unruh relation binding acceleration to temperature through a constant the dimensions demand.

Key consequence

The one temperature in the basis is zero for inertial observers and proportional to acceleration for the rest. The thermal and kinematic quantities are tied together through the vacuum.

7 A basis, and a question

The reconstruction leaves dimensional analysis looking less like bookkeeping than it usually appears. The choice of base dimensions is not a physical question about which quantities the world permits us to regard as fundamental, and the answer changed in 1905 without the practice of dimensional analysis registering the change.

A post-relativistic basis keeps the interval, which fixes c and merges length with time; proper acceleration, the kinematic invariant that outlives velocity; and thermal energy, which supplies mass and dispenses with Boltzmann's constant. Planck's constant enters as the forced bridge between acceleration and energy, with the Unruh effect setting its coefficient and explaining why the pairing is canonical. Two invariant quantities do the work of Maxwell's three. Whether physics ought to adopt such a basis is a separate matter, and not the claim made here. Dimensional analysis is usually presented as bookkeeping. Its bookkeeping, however, begins by declaring certain quantities fundamental. Maxwell's choices reflected the physics of 1873. Einstein changed the ontology of space and time in 1905. The formal machinery of dimensional analysis incorporated neither the interval nor proper acceleration into its foundations. Whether or not one adopts the reconstruction proposed here, the episode illustrates that dimensional bases are more than conventions of notation being historical claims about what physics regards as primitive. See related mass and entropy scale discussion [6] and the cosmological consequences of a Hubble-adapted basis [7]

References

- [1] J. C. Maxwell, *A Treatise on Electricity and Magnetism*, vol. 1, Clarendon Press, Oxford (1873), preliminary chapter, “On the Measurement of Quantities.”
- [2] E. Buckingham, “On Physically Similar Systems; Illustrations of the Use of Dimensional Equations,” *Phys. Rev.* **4**, 345–376 (1914).
- [3] P. W. Bridgman, *Dimensional Analysis*, Yale University Press, New Haven (1922).
- [4] W. G. Unruh, “Notes on black-hole evaporation,” *Phys. Rev. D* **14**, 870–892 (1976).
- [5] Bureau International des Poids et Mesures, *The International System of Units (SI)*, 9th ed. (2019); definition of the kelvin by fixing k_B .
- [6] L. McCulloch-James, “How Many Scales Can You Build? Dimensional Analysis, Gaussian Elimination, and the Geometry of Mass and Entropy,” ResearchGate preprint (2026). doi: [10.13140/RG.2.2.21404.76165](https://doi.org/10.13140/RG.2.2.21404.76165).
- [7] L. McCulloch-James, “From Hubble Action to Horizon Entropy: One Cosmological Hierarchy in Two Dimensional Bases,” ResearchGate preprint (2026).